Session 3 (2h) — ADC/PWM & 센서 연동

Covers: **ADC/PWM**, **센서 연동** (on-site task)

1. ADC — Analog-to-Digital Conversion — ~35 min

- Digital pins only see HIGH/LOW. Real-world sensors (light, temperature, potentiometers) output a *continuous* voltage — the ADC converts that to a number the MCU can use.
- Arduino Uno's ADC is 10-bit → `analogRead(pin)` returns **0–1023** for 0–5V on pins A0–A5.
- This is literally how you do **센서 연동 (sensor integration)**: almost every basic analog sensor (potentiometer, LDR/photoresistor, simple temp sensor) is read this way.

```
int sensorValue = analogRead(A0); // 0-1023
```

2. PWM — Pulse Width Modulation — ~35 min

- Digital pins can only be fully on or off, but `analogWrite(pin, value)` rapidly switches a pin on/off so fast it *behaves* like a variable analog voltage — used for LED dimming, motor speed, servo-style control.
- `value` range: 0 (off) – 255 (fully on).
- **Only works on `~` marked pins** (Uno: 3, 5, 6, 9, 10, 11) — these have dedicated timer hardware for it. This is a common beginner mistake (using a non-PWM pin) — check the board silkscreen for the `~`.

3. Connecting ADC → PWM (real sensor integration pattern) — ~30 min

The actual 현장 업무 pattern: read a sensor value with `analogRead()`, then use that value to drive an output with `analogWrite()` — e.g. brighter room → dimmer LED, or a potentiometer directly controlling LED brightness. This is the core loop behind most sensor-driven embedded projects.

```
int sensorValue = analogRead(A0);        // 0-1023
int pwmValue = map(sensorValue, 0, 1023, 0, 255); // rescale to PWM range
analogWrite(9, pwmValue);
```

Practice: `practice/sensor_pwm_control.ino` — ~20 min

A potentiometer (or any analog sensor) on A0 directly controls an LED's brightness on D9 via PWM — the ADC→PWM pattern above, working end to end.

Wiring for this session

Component	Pin	Other end
Potentiometer wiper (middle pin)	A0	—
Potentiometer outer pins	5V and GND	one to each
LED (+)	D9 (~PWM)	220Ω resistor → anode
LED (–)	GND	cathode

> No potentiometer in your kit? In Tinkercad, drag one in for this exercise — it's free and this ADC↔PWM pattern is the single most reusable skill for the internship's "센서 연동" task, worth practicing in simulation even without the physical part.

Checkpoint before Session 4

- [ ] Can explain what `analogRead()` returns and its range
- [ ] Can explain what `analogWrite()` does and which pins support it
- [ ] LED brightness follows the sensor/potentiometer smoothly